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EMG differences between concentric and eccentric maximum voluntary contractions are evident prior to movement onset

Received: 28 August 2001 / Accepted: 5 March 2002 / Published online: 28 May 2002
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Abstract This investigation addressed the question of whether the muscle activation signal prior to movement onset, as measured by surface EMG, differs if the contraction to be performed is concentric (shortening) or eccentric (lengthening). Specifically, the purpose was to determine if differences in knee extensor muscle EMG prior to voluntary maximum concentric and eccentric contractions and initiated from the same knee joint angle are evident at a time before muscle length changes could be influential. A protocol was designed using isokinetic knee extensions. The EMG of the vastus lateralis, vastus medialis, rectus femoris, and hamstrings muscles and the associated knee extension moment were measured during the isometric phase preceding the onset of dynamometer motion. During this isometric phase the muscles initially contracted under identical conditions, irrespective of whether the contraction was to be concentric or eccentric. The EMG of the eccentric contractions was significantly smaller than that of the concentric contractions. However, the rate of change of knee extension moment generally did not differ between the two conditions. This was found for both the monoarticular and multiarticular knee extensor muscles. The results suggest that initial differences between the EMG of maximum voluntary concentric and eccentric knee extensor contractions are selected *a priori* and support the contention that the central nervous system distinguishes between maximum eccentric and concentric contractions. The emergence of differences in activation prior to muscle length changes suggests supraspinal influences.

Keywords Electromyography · Isokinetic · Muscle · Motor control

Introduction

Accumulating evidence suggests that the central nervous system (CNS) uses different control strategies for eccentric (lengthening) contractions and concentric (shortening) contractions (Enoka 1996). This is commonly observed, indirectly, as a general inability of the CNS to fully activate the motoneuron pool during maximum voluntary eccentric contraction (Westing et al. 1991). Compared to a concentric maximum voluntary contraction (MVC), an eccentric MVC is usually associated with a lower activation level as measured by surface EMG. However, some authors have reported that under some conditions, activation levels during concentric and eccentric contractions are not different (Komi et al. 2000). Incomplete activation during eccentric MVCs of the knee extensor muscles was demonstrated by a significant increase in knee extension moment when electrical stimulation of the knee extensor muscles was superimposed over voluntary effort (Westing et al. 1990). In contrast, the isometric and concentric MVCs were not influenced by superimposing electrical stimulation over the voluntary effort. It is possible that the differences in the ability to fully activate the knee extensor muscles are dependent on the speed at which the muscles contract (Babault et al. 2001). However, the contraction speed dependence of activation may be muscle specific (Pinniger et al. 2000).

The differences in EMG between concentric and eccentric contractions are further reflected by differences in the excitability of the involved spinal motor neurons (Nakazawa et al. 1997, 1998; Pinniger et al. 2001) and by observations of some task-dependent supraspinal influence on the motor neuron pool (Abbruzzese et al. 1994). In addition, there is evidence, derived from EEG studies, that cortical activity during the planning for concentric and eccentric contractions differ (Fang et al. 2001).

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A second line of evidence suggesting incomplete activation during eccentric MVC is the difference in the time course of fatigue demonstrated by human subjects performing maximum eccentric or maximum concentric MVCs of the knee extensor muscles. Repeated isokinetic knee extension MVCs produce significant reductions in concentric MVC but much smaller reductions in eccentric MVC (Gray and Chandler 1989; Tesch et al. 1990; Crenshaw et al. 1995; Grabiner and Owings 1999). Similar observations have been reported for plantarflexion contractions (Hortobágyi et al. 1996) and dorsiflexion contractions (Pasquet et al. 2000).

It is possible that the time course of fatigue for concentric and eccentric contractions would more approximate one another if human skeletal muscle could be voluntarily fully activated during maximum voluntary eccentric contractions. For example, the increased knee extensor strength following a training protocol was associated with increased levels of voluntary activation and fatigability (Hortobágyi et al. 1996). This finding links a training-induced increase in the level of knee extensor muscle activation and muscle fatigue during eccentric MVCs.

The underlying neural and mechanical mechanisms that give rise to activation differences in concentric and eccentric contractions, as well as the interactions between them, have not been fully characterized. Ultimately, these mechanisms converge on the motoneuron pool and influence motor unit discharge rate and/or motor unit recruitment. Indeed, for similar external joint moments, the discharge rate of motor units is lower during submaximum eccentric contractions compared to submaximum concentric contractions (Tax et al. 1989, 1990; Kossev and Chritova 1998). In addition, there is limited evidence suggesting that submaximum eccentric contractions may employ a motor unit activation strategy favoring high threshold motor units (Nardone and Schieppati 1988; Nardone et al. 1989; Howell et al. 1995). Because of the technical difficulty of tracking motor unit behavior during high intensity contractions, it remains an open question as to whether MVCs are associated with similar discharge patterns. Overall, however, there is a dearth of direct evidence that concentric and eccentric contractions are controlled differently by the CNS.

In the present work, two experiments using a novel experimental protocol and analysis addressed the issue of differences in the initial muscle activation, as measured by EMG, during the isometric phase preceding maximum voluntary concentric and eccentric contractions. The purpose of the first experiment was to determine if differences in the initial EMG of vastus lateralis during voluntary maximum concentric and eccentric contractions initiated from the same knee joint angle are evident prior to the time at which muscle length changes could be influential. The second experiment extended the protocol to include simultaneous measurement of multiple quadriceps femoris components. Broadly, we have asked the question "Does the activation signal that reaches the knee extensor muscle in-

clude information regarding the type of contraction to be performed?".

Materials and methods

Subjects

Seven women (age: 22.0 ± 1.5 years; height: 1.63 ± 0.06 m; weight: 626.2 ± 123.0 N) and 17 men (age: 25.9 ± 3.5 years; height: 1.80 ± 0.08 m; weight: 773.0 ± 126.1 N) participated in experiment 1. For experiment 2, 3 women (age: 24.0 ± 4.5 years; height: 1.61 ± 0.02 m; weight: 561.9 ± 29.6 N) and 6 men (age: 27.2 ± 5.6 years; height: 1.76 ± 0.06 m; weight: 757.1 ± 157.8 N) participated. All subjects were healthy, recreationally active, and reported having no history of injury to their right knee. One woman participated in both experiments. The methods were reviewed and approved institutionally and subjects provided informed consent prior to participation.

Dynamometer

Maximum voluntary contractions of the right knee extensor muscles were performed on a Kin-Com isokinetic dynamometer (Chattanooga Corporation, Chattanooga, Tenn., USA). The contractions were performed at a nominal isokinetic speed of 60° s^{-1} . All contractions, both concentric and eccentric, were initiated with the knee joint initially positioned at 45° of flexion (zero degrees flexion represents full knee extension). The range of motion for both concentric and eccentric contractions was 30° . During the protocol the subject was seated upright. The subject's waist and distal right thigh were stabilized using Velcro straps. The thigh-trunk angle was fixed at approximately 100° . The dynamometer axis of rotation was aligned with the putative sagittal plane center of rotation of the knee joint. The dynamometer load cell is uniaxial and measures the compressive force applied perpendicular to its surface (i.e., the load cell does not measure moment). The load cell was positioned over the leg proximal to the ankle joint and secured with a Velcro strap.

The signal from the load cell provided the trigger to activate the dynamometer servomechanism and initiate dynamometer motion. A 40-N load, which is the instrument default "threshold" value, was required to trigger the servomechanism. The magnitude of the threshold value influences the duration of the period between the onset of muscle activation and the onset of dynamometer motion, during which the muscle is effectively contracting isometrically. In experiment 1 the threshold was 40 N for all subjects. In experiment 2, a subject-specific threshold, equal to 30% of each subject's isometric MVC, was used. The group mean and standard deviation of the threshold was 203.7 ± 43.04 N. By increasing the threshold, the duration of the isometric period was increased. This threshold value was determined in a pilot collection. In this collection, we systematically increased the threshold to determine the largest value that could be used before the subject could distinctly sense the transition between the increase of isometric knee extension moment, the "quick-release" of the dynamometer, and dynamic phase of the ensuing contraction.

Electromyography

For both experiments, EMG was recorded using bipolar silver-silver chloride electrodes (recording diameter=8 mm; center-to-center distance=20 mm). The electrodes were placed over the vastus lateralis and hamstrings muscles of the right leg in the assumed direction of the underlying muscle fibers. The hamstrings electrodes were placed at the midpoint of a line running along the center of the posterior aspect of the thigh from the gluteal fold to the knee joint center. A reference electrode was placed over the fibular head. For experiment 2, in addition to the vastus lateralis and ham-

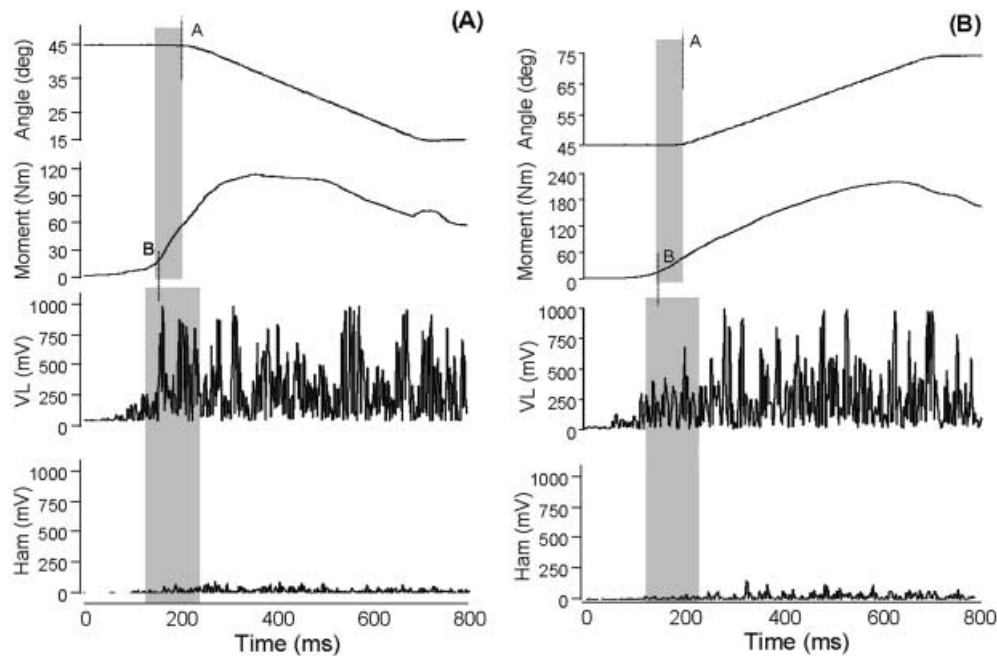


Fig. 1 **A** Data from a single concentric trial from experiment 1 showing the knee angle, knee extension moment, and rectified EMG of vastus lateralis (VL) and hamstrings (Ham) as a function of time. The threshold force, necessary to activate the dynamometer servomechanism and initiate dynamometer motion, was 40 N. The onset of dynamometer motion (dotted line A) determined the endpoint of the 50-ms analysis window (shaded region of knee position and knee extension moment) used to compute the rate of change of knee extension moment. The location of the vertical dotted line B represents the time at which 5% maximum voluntary contraction (MVC) of the knee extension moment was generated. The 100-ms window (shaded region of VL), which began 25 ms before line B, was used to compute the rectified and integrated EMG of the VL and Ham muscles. **B** Data from a single eccentric trial from experiment 1

strings, vastus medialis and the rectus femoris EMG were recorded using similar surface electrode configurations. The measured interelectrode resistance in most cases was less than 5 and 10 k Ω for men and women, respectively.

The detected EMG signals were amplified (Noraxon, Scottsdale, Ariz, USA; gain=1,000, CMRR=130 dB, bandpass=15–500 Hz), digitized at 1 kHz (Data Translation 2801A), and stored for post-processing. The force, angular velocity, and angular position signals from the dynamometer were similarly digitized.

Protocol

In experiment 1, each subject performed a block of three trials of concentric MVCs and a block of three trials of eccentric MVCs. The order in which the blocked trials were performed was randomized. For experiment 2, a block of concentric MVCs and a block of eccentric MVCs, ten trials per condition, were performed in a randomized order. For both experiments, 1 min of rest was given between MVCs to reduce fatigue. Subjects were instructed which type of contraction would be performed and to perform the knee extension MVCs through the entire range of motion following a verbal “Go” signal. Verbal encouragement was vigorously provided throughout the entire trial.

Data analysis

The digitized knee extension force data were smoothed using a recursive, lowpass Butterworth filter (cut-off frequency=10 Hz) and converted to knee extension moment values. Using custom graphics software, the onset of dynamometer motion was visually determined by the investigator as an observable deflection of the angular position signal from the dynamometer (Figs. 1, 2 upper panels).

For each trial of experiment 1, a 50-point (i.e., 50 ms) linear regression was used to characterize the knee extension moment as a function of time. The first point in the time series used by the linear regression was 50 points prior to the instant identified as the onset of the dynamometer motion. The slope coefficient from the computed linear regression equation was used as a dependent variable representing the initial rate of change of isometric knee extension moment (dM/dt). The period during which dM/dt is computed occurs prior to the onset of dynamometer motion. In experiment 1 the analysis windows for quantifying the EMG were similar, but not identical, to the windows used to analyze dM/dt . A 100-point time series was used, the first point of which was 25 ms prior to the instant at which the knee extension moment exceeded 5% of the peak value (Fig. 1). The EMG signal was full-wave rectified, integrated, and averaged across trials.

In experiment 2, the positioning of the analysis windows for dM/dt and EMG was changed from those of experiment 1. The change allowed the use of the same window for both dM/dt and EMG. This was accomplished by extending the duration of the isometric period preceding the onset of dynamometer motion by increasing the threshold required to initiate dynamometer motion. This allowed the initial point of the analysis window to be positioned at 100 ms prior to the onset of dynamometer motion (Fig. 2). Similar to experiment 1, although there is internal shortening of the active muscles, the dM/dt and EMG variables may be considered to have occurred during an *isometric condition*, regardless of the contraction type performed for that trial.

Statistics

Paired *t*-tests were used to compare the rate of change of knee extension moment and the initial EMG for the concentric and eccentric MVCs. Statistical analyses were performed using SPSS Version 7.0. Statistically significant differences were identified using a *P* value of 0.05.

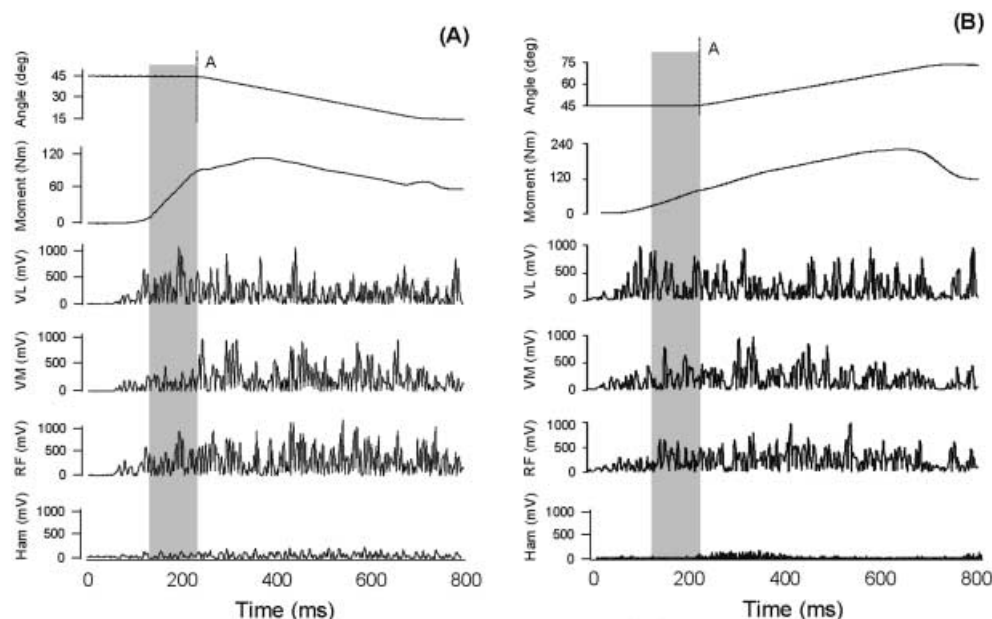


Fig. 2 **A** Data from a single concentric trial from experiment 2 showing the knee angle, knee extension moment, and rectified EMG of vastus lateralis (VL), vastus medialis (VM), rectus femoris (RF), and hamstrings (Ham) as a function of time. The threshold force, necessary to activate the dynamometer servomechanism and initiate dynamometer motion, was the subject-specific 30% MVC. Onset of dynamometer motion (dotted line A) determined the endpoint of the 100-ms analysis window (shaded region). The data within this window was used to compute the rate of change of knee extension moment and the rectified and integrated EMG of the VL, VM, RF, and Ham muscles. **B** Data from a single eccentric trial from experiment 2

Results

The rate at which knee extension moment was initially generated was not affected by the concentric and eccentric contraction conditions. In experiment 1, the difference between dM/dt preceding the onset of eccentric and concentric dynamometer motion was not significant ($P=0.969$; Table 1). In experiment 2, the difference between dM/dt for the concentric and eccentric conditions did not achieve significance ($P=0.056$, computed statistical power=0.5; Table 1).

Table 1 The slope coefficient (dM/dt in N m/ms) and the initial activation level (mV) for the concentric and eccentric contractions of experiment 1 and experiment 2, and the percent change be-

	Experiment 1 (n=24)			Experiment 2 (n=9)		
	Concentric	Eccentric	Δ	Concentric	Eccentric	Δ
dM/dt	0.93 (0.43)	0.93 (0.35)	0	0.95 (0.26)	0.86 (0.27)	-9.5
VL	333.5 (231.9)	248.9 (157.7)*	-25.4	348.2 (72.1)	303.4 (70.6)**	-12.9
VM	—	—	—	427.9 (113.1)	365.7 (101.0)**	-14.5
RF	—	—	—	297.8 (66.4)	261.7 (56.1)**	-12.1
HAM	27.0 (9.7)	26.5 (14.0)	-1.9	38.1 (18.1)	34.7 (13.2)	-8.9

* $P<0.05$

** $P<0.01$

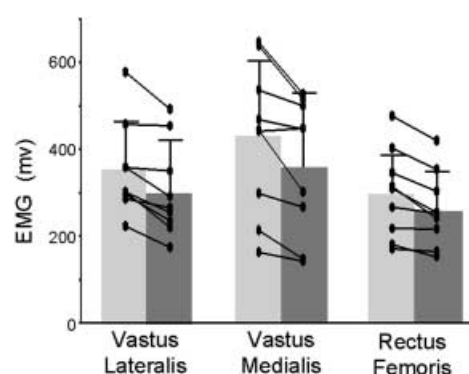


Fig. 3 Mean, standard deviation, and subject-specific rectified and integrated EMG of the vastus lateralis, vastus medialis, and rectus femoris during concentric (light gray) and eccentric (dark gray) contractions during experiment 2

Compared to the concentric contraction conditions, the initial EMG was diminished during the eccentric contraction conditions. For experiment 1 and experiment 2, the initial EMG of the vastus lateralis during the eccentric MVC was significantly smaller than that of

tween the contractions [$\Delta=100 \times (\text{eccentric} - \text{concentric}) / \text{concentric}$]. Values are mean (standard deviation). (VL vastus lateralis, VM vastus medialis, RF rectus femoris, HAM hamstrings)

the concentric MVC ($P=0.035$ and 0.002 , respectively; Table 1). For experiment 2, the initial EMG of the vastus medialis and the rectus femoris during the eccentric MVC were significantly smaller than that of the concentric MVC ($P=0.009$ and 0.003 , respectively; Table 1; Fig. 3).

The results of dM/dt and initial EMG level were not a function of altered antagonist coactivation. The levels of hamstrings coactivation during the eccentric and concentric conditions were not significantly different for experiment 1 or for experiment 2 ($P=0.817$ and 0.336 , respectively; Table 1).

Discussion

The objective of the study was to address the question, "Does the activation signal that reaches the muscle include information regarding the type of contraction to be performed?". By using a novel protocol and analysis method, the differences in activation level, as measured by EMG, and force generation during concentric and eccentric contractions could be evaluated without the influence of muscle shortening and lengthening, respectively.

The results from experiment 1 show that the EMG of the vastus lateralis during the isometric phase preceding eccentric contractions was lower than that preceding the concentric contractions even though the between-condition differences in the rate of knee extension moment were not significant. Experiment 2 confirmed these results and also revealed that the vastus medialis and rectus femoris showed a similar decrease in EMG during the isometric phase preceding eccentric contractions. Coactivation of the antagonist hamstrings is not considered to have influenced the results since the between-condition difference in hamstrings EMG was not significant.

Experiments 1 and 2 differed qualitatively with respect to the extent that activation level was reduced during the eccentric contraction condition. In experiment 1 the activation level of the vastus lateralis was reduced by about 25% during the eccentric contraction condition whereas in experiment 2 the activation level of the vastus lateralis was reduced by about 13%. In experiment 2 the activation levels of the vastus medialis and rectus femoris mimicked that of vastus lateralis. Furthermore, in experiment 2 the dM/dt was qualitatively, but not significantly, lower during the eccentric contraction condition than the concentric contraction condition. Given that the experimental methods were similar, we computed the harmonic means and harmonic standard deviations for the dM/dt and vastus lateralis activation level during the concentric and eccentric contraction conditions. The harmonic values provide descriptive data for an experiment having a sample size of 33 subjects. For dM/dt , the difference between the concentric and eccentric contraction conditions (0.94 ± 0.45 and 0.91 ± 0.39 , respectively) was -3.2% . For the vastus lateralis activation level, the difference between the concentric and eccentric contraction conditions (337.5 ± 235.9 and 263.8 ± 165.7 , respectively)

was -21.8% . These values, which more closely resemble those of experiment 1 ($n=24$) than experiment 2 ($n=9$), suggest that the qualitative between-experiment differences may have been a function of the sample size.

The findings of these experiments are thought provoking and raise the question, "How can two contraction conditions be associated with similar dM/dt have significantly different activation levels?". The protocol was designed such that conditions from which the concentric and eccentric contractions were initially performed were similar; the contractions were initiated from the same knee joint angle, the force threshold, tantamount to the knee extension moment required to initiate dynamometer motion, was identical in the concentric and eccentric conditions for each subject, and the between-condition differences in antagonist hamstrings muscles coactivation were not significant. The answer to the above question should ultimately be associated with between-condition differences in motor unit behavior. However, the present study relied on the use of surface electrodes that precludes the characterization of motor unit recruitment and rate coding. However, the methodologies that are amenable to characterization of motor unit behavior, i.e., intramuscular electrodes and EMG signal decomposition techniques, may offer challenging technical difficulties for large locomotor muscles contracting at very high levels of MVC.

A feasible approach to interpreting the results of the present study relative to the underlying motor behavior must couple experimentally derived data with a mathematical model of a motor unit pool. Such a model (see, for example, Fuglevand et al. 1993) can facilitate exploring motor unit recruitment and rate coding patterns that could give rise to the force-EMG patterns derived experimentally. It does not seem likely that orderly recruitment of motor units could result in similar dM/dt and significantly lower activation levels. Selective activation of large motor units would be expected to increase dM/dt to an extent somewhat proportional to the numbers of small motor units that are not activated. The surface EMG of such an activation pattern would be expected to decrease. A valid motor unit model will provide a platform by which very specific activation patterns could be systematically explored. Such a model, applied to the present data set could elucidate the extent to which the present results are consistent with a control strategy in which large motor units were, to some extent, selectively activated.

A novel aspect of the current study is that comparisons between concentric and eccentric contractions were made prior to the onset of dynamometer motion at identical knee joint angles. Feedback from muscle and joint mechanoreceptors signaling muscle length and joint position at the *beginning* of each trial could be assumed to be similar for concentric and eccentric contraction conditions. Furthermore, comparisons made prior to the onset of dynamometer motion precludes the need to consider the influence of peripherally derived feedback signaling muscle shortening, during the concentric contractions, and muscle lengthening, during the eccentric contrac-

tions. These aspects of the experimental protocol increase the confidence that the results reflect a change in the control strategy rather than a postmovement modification of the control strategy.

The analysis windows for dM/dt and for initial activation level in experiments 1 and 2 were different due to differences in the thresholds used. In experiment 1, it was observed during data processing that the elapsed time between initial moment generation and the onset of dynamometer motion was found to be less than 100 ms for some subjects. This was attributed to the use of the 40-N threshold. To accommodate, the analysis window for determining dM/dt was reduced to 50 ms in order to remain within the isometric phase of the trial and to diminish the influence of the non-linear portion of the dM/dt curve prominent at the beginning of the time series. A 100-ms analysis window for the EMG was retained but the first point of the window was relocated to coincide with the time at which 5% MVC of the knee extension moment curve occurred. However, for some trials this 100-ms analysis window extended beyond the isometric phase preceding dynamometer motion. Nevertheless, the extent to which all of these trials extended beyond the isometric phase was always smaller than the latency for a spinal reflex. Therefore, the possibility that changes in muscle length influenced the quantified EMG was discarded. Nevertheless, this consideration was eliminated in experiment 2 by increasing the force threshold level to 30% of the subjects' isometric MVC. This was sufficient to allow a 100-ms analysis window for both the rate of moment generation and muscle activation to be within the isometric phase.

Although the vastus lateralis EMG from experiment 1 was significantly reduced during the eccentric contractions, we could not discount the possibility that EMG of other quadriceps femoris components was increased to accommodate this reduction. Thus in experiment 2, the multiarticular rectus femoris and the uniarticular vastus medialis were also examined. The results showed that the reduction in EMG occurred in each of the quadriceps femoris components tested. However, one quadriceps femoris component, vastus intermedius, is not accessible with surface electrodes. Therefore, we cannot be certain that vastus lateralis activation is consistent with the patterns of the measured quadriceps components. This presently remains an open question.

The finding of significant differences between the initial EMG during concentric and eccentric MVCs provides experimental support for the contention that the CNS implements a different motor command for eccentric MVCs. The results of the present study are novel, however, because the differences between concentric and eccentric MVCs relative to activation level and moment-generation were evident before changes in muscle length could have influenced the results. This implies that the activation signals were independent of muscle length changes and thus, independent of feedback from the periphery. The results suggest that the specific activation

signals were likely a function of the volitional intent of the imminent motion.

Acknowledgements The authors would like to express their gratitude to Michelle George, Alex Hamilton, and Kristy Ebinger who assisted during the data collection and analysis, and to the Chattanooga Corporation for their partial support of this work.

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